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MOVW A,&A+DPTR
JZ OVER
MOV BUF,A
LCALL DAT
INC DPTR
SJMP HERE1

OVER:   LCALL ONE_SEC_DELAY
        LCALL ONE_SEC_DELAY
        LCALL CLEAR
        MOV BUF,&000H
        LCALL CMD
        MOV DPTR,&MSG7
HERE2:  CLR A
        MOVW A,&A+DPTR
        JZ CONVERT
        MOV BUF,A
        LCALL DAT
        INC DPTR
        SJMP HERE2

CONVERT:LCALL DDELAY
CLR CS ; INITIATE CONVERSION
SETB CLK
CLR CLK ; FIRST CLOCK
SETB CLK
CLR CLK ; SECOND CLOCK
MOV A,&00H ; CLEAR A
MOV R5,&08H ; 8 CLOCK PULSES
AGAIN: MOV C,DO
RLC A
SETB CLK
CLR CLK
DJNZ R5,AGAIN
SETB CS
MOV ADC_VAL,A
MOV B,&79D
MUL AB ; PRODUCT IN AB
MOV R1,B ; HIGH BYTE IN B
MOV R2,A ; LOW BYTE IN A
LCALL HEX2BCD
        MOV VAL1,R7
        MOV VAL2,R6
        MOV VAL3,R5
        LCALL SENDVAL2LCD

CHECK:  JNB LBF,CHECK2 ; SEE IF ANY
        FLAGS ARE SET ,i.e. LOW BATT FLAG OR BATT
        FULL FLAG
        JB DYI,CONVERT
CHECK2:JNB OCF,PROCEED
        JB DYI,NIGHT
        SJMP CONVERT
PROCEED: JB DYI,NIGHT
        CLR LD_RL ; OFF LOAD
        CLR LBF ; CLEAR LOW BATT FLAG
        MOV A,VAL2 ; SEE IF BATT.IS FULL
        XRL A,&04H
        JZ FULLCHG
        CLR CHG_RL ; CONNECT BATT. TO
        PANEL
        MOV DPTR,&MSG4 ; DISPLAY CHARGING
        MSG
        MOV IMS,&00H
        LCALL SENDSTAT2LCD
        LJMP CONVERT
FULLCHG: SETB OCF ;SET OVERCHARGE
        SETB CHG_RL ;DISCONNECT BATT.FROM
        PANEL
        MOV TH0,&03CH ;START 5 MIN TIMER
        HERE
        MOV TL0,&0B0H ;DISCONNECT BATT FROM
        PANEL
        MOV CNT1,&200D
        MOV CNT2,&30D
        SETB ETO
        SETB TRO
        SETB EA
        MOV DPTR,&MSG5 ; DISPLAY BATT.FULL
        MSG

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MOV IMG,#001H
LCALL SENDSTAT2LCD
LJMP CONVERT
NIGHT: CLR CHG_RL ; RECONNECT BATT.
                TO PANEL
CLR TR0 ; STOP TIMER0 INCASE ITS
                RUNNING
CLR OCF ; CLEAR OVER CHARGE FLAG
SETB LD_RL ; CONNECT LOAD TO BATT.
MOV A,VAL1
XRL A,#00H
JZ LOWBAT
MOV DPTR,#MSG3 ; DISPLAY LOAD ON
MSG
MOV IMG,#002H
LCALL SENDSTAT2LCD
LJMP CONVERT
LOWBAT: SETB LBF
CLR LD_RL ; DISCONNECT LOAD FROM
                BATT.
MOV DPTR,#MSG6 ; DISPLAY BAT.OV AND
                LOAD OFF MSG

MOV IMG,#003H
LCALL SENDSTAT2LCD
LJMP CONVERT
SENDVALLCD: MOV BUF,#0C7H
LCALL CMD
MOV A,VAL1
ORL A,#30H
MOV BUF,A
LCALL DAT
MOV A,VAL2
ORL A,#30H
MOV BUF,A
LCALL DAT
MOV BUF,#7.'
LCALL DAT
MOV A,VAL3
ORL A,#30H
MOV BUF,A
LCALL DAT
RET
SENDSTAT2LCD: MOV BUF,#080H
LCALL CMD
HERE3: CLR A
MOVC A,#A+DPTR
JZ PICT
MOV BUF,A
LCALL DAT
INC DPTR
SJMP HERE3
PICT: MOV BUF,#0CEH
LCALL CMD
MOV BUF,IMG
LCALL DAT
BACK: RET
;*****
; TIMER0 ISR (5 MINUTES TIMER)
;*****
COUNTDOWN: CLR TR0
MOV TH0,#03CH
MOV TLO,#0B0H
SETB TR0
DJNZ CNT1,BACK2
MOV CNT1,#200D
DJNZ CNT2,BACK2
CLR TR0 ; OFF 5 MIN TIMER
CLR ET0
CLR OCF ; CLEAR OVER CHARGE FLAG
CLR CHG_RL ; RE-CONNECT BATT TO
                PANEL
BACK2: RETI
Hex2BCD: MOV R3,#00D
MOV R4,#00D
MOV R5,#00D
MOV R6,#00D
MOV R7,#00D
ACALL H2B

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RET
R2B:    MOV B,#10D
MOV A,R2
DIV AB
MOV R3,B ;
MOV B,#10 ; R7,R6,R5,R4,R3
DIV AB
MOV R4,B
MOV R5,A
CJNE R1,#00H,HIGH_BYTE ; CHECK FOR HIGH BYTE

        SJMP ENDD
HIGH_BYTE:  MOV A,#6
            ADD A,R3
            MOV B,#10
            DIV AB
            MOV R3,B
            ADD A,#5
            ADD A,R4
            MOV B,#10
            DIV AB
            MOV R4,B
            ADD A,#2
            ADD A,R5
            MOV B,#10
            DIV AB
            MOV R5,B
            CJNE R6,#00D,ADD_IT
            SJMP CONTINUE
ADD IT:   ADD A,R6
CONTINUE:MOV R6,A
DJNZ R1,HIGH_BYTE
MOV B , #10D
MOV A,R6
DIV AB
MOV R6,B
MOV R7,A
ENDD: RET
ONE_SEC_DELAY: MOV R0,#10D ; One second delay routine
                MOV R1,#100D
R23: MOV R1,#100D
R21: MOV R2,#250D
R22: NOP
      NOP
      DJNZ R2,R22
      DJNZ R1,R21
      DJNZ R0,R23
RET
PWR_DELAY: ; 15 mSec DELAY FOR LCD TO INITIALIZE AFTER POWER-ON
MOV R4,#100D
H2: MOV R3,#250D
H1: DJNZ R3,H1
     DJNZ R4,H2
     RET
;*****LCD SUBROUTINES*****
CMD: PUSH ACC ; SAVE ACCUMULATOR
SETB EN
CLR RS ; SELECT SEND COMMAND
MOV A,BUF ; PUT DATA BYTE IN ACC
MOV C,ACC.4 ; LOAD HIGH NIBBLE ON DATA BUS
MOV DB4,C ; ONE BIT AT A TIME USING...
MOV C,ACC.5 ; BIT MOVE OPERATOINS
MOV DB5,C
MOV C,ACC.6
MOV DB6,C
MOV C,ACC.7
MOV DB7,C
CLR EN
NOP
SETB EN ; PULSE THE ENABLE LINE
MOV C,ACC.0 ; SIMILARLY, LOAD LOW NIBBLE
MOV DB4,C
MOV C,ACC.1
MOV DB5,C

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